

Autodesk® Scaleform®

Scaleform

Scaleform 4.0

: Matthew Doyle
:
:
: 2011 4 25

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Autodesk® Scaleform® 4.3

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:

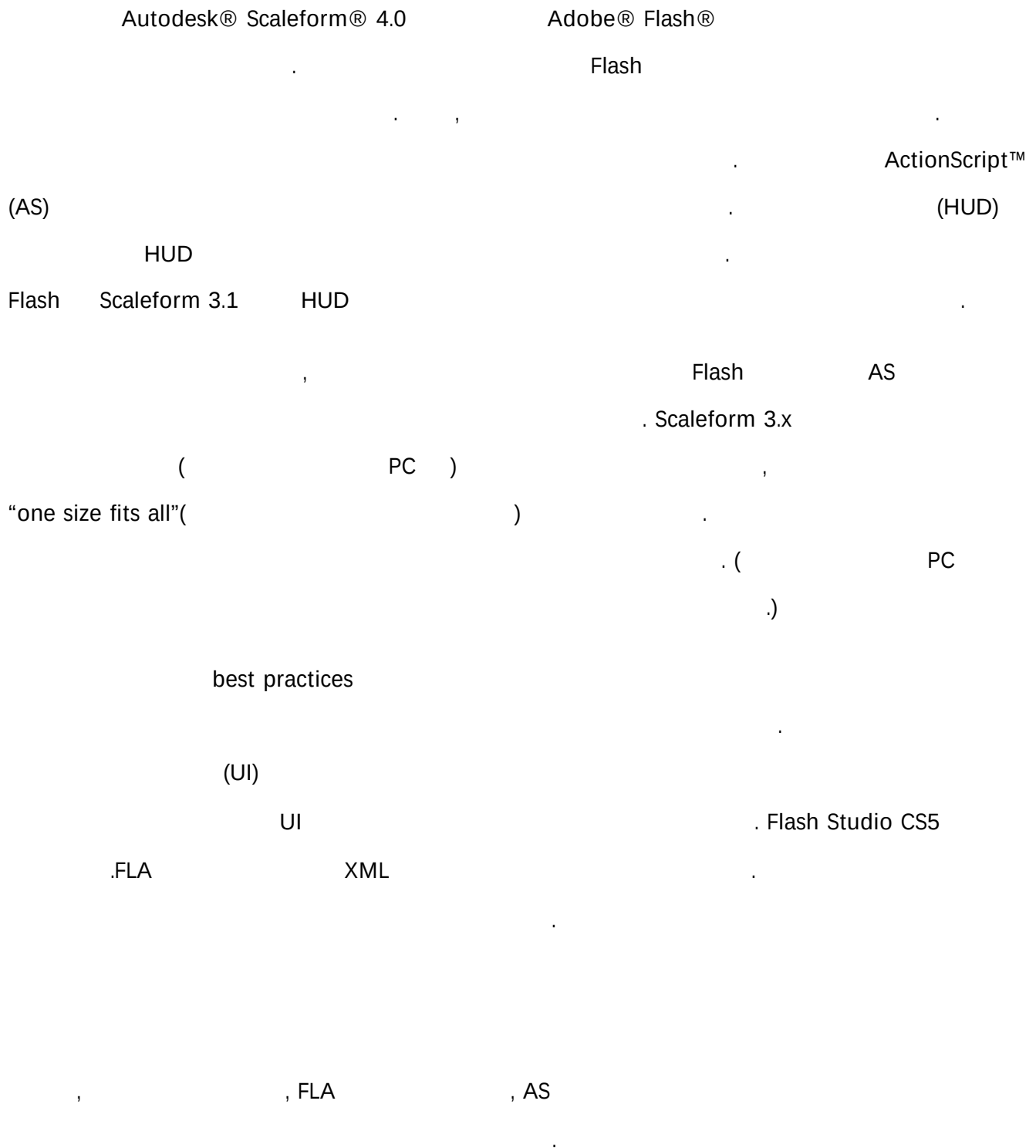
	Scaleform Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA www.scaleform.com info@scaleform.com (301) 446-3200 (301) 446-3199
--	---

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1 개요



- Flash AS :
- 0

- FLA AS :
- 0 AS , ,

- :
- 0

- :
- 0

best practice

Flash

-
-

(scope)

1.1

-

- () UI

Scaleform 3.3

Scaleform Common Lightweight Interface Kit (CLIK™)

1.2

-

(transform)

-

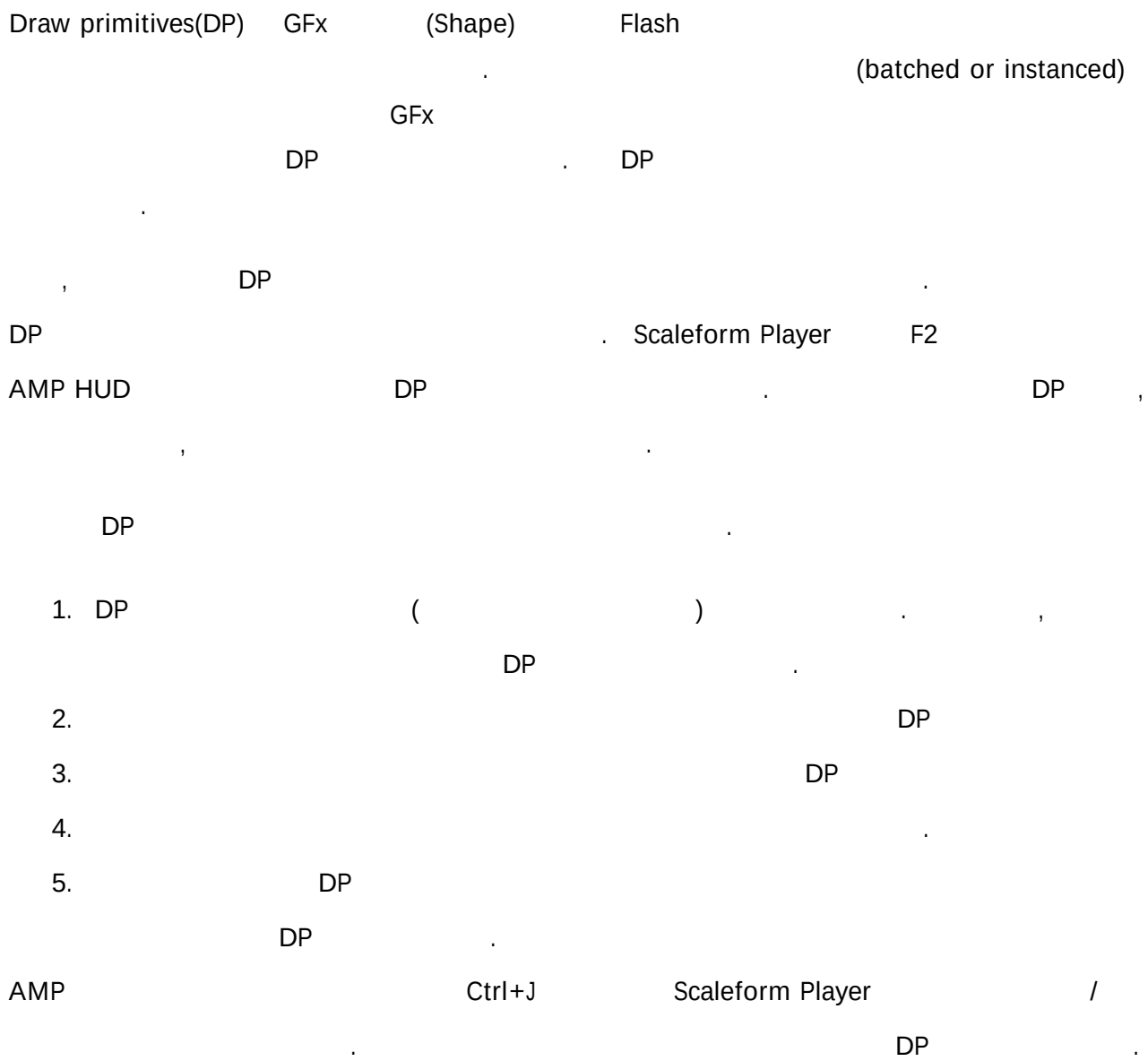
Flash

, MMORPG

- HUD
 - AS SWF AS2 AS3 API
- - SWF C++ API load unload
 - (Gfx::Movie .)
- Scaleform 3.3
 - CPU
 - CD DVD
- SWF
- UI
 - o HUD : 400K–1.5M
 - o / : 800K–4M
 - o (Pacman): 700K–1.5M
 - o : 2M–10M+

2 메모리에 콘텐츠를 생성하는 것과 성능상의 고려사항

2.1 Draw Primitives



2.2

1.

Advance

2.

(nesting)

3.

`_alpha=0` `_visible=false` (in AS2) and
`visible=false` rather than `alpha=0` (in AS3).

`stop()`

2.3 Artwork

2.3.1 비트맵 vs 벡터 그래픽

Scaleform

2.3.2 백터 그래픽

()

$$\left(\begin{array}{c} \text{ } \\ \text{ } \end{array} \right)$$

-

0 (solid fills)

0 ()

alpha(AS2) or alpha (AS3)

```
0      _visible(AS2) or visible (AS3)      false
```

_alpha/alpha property 100

()

. x y(in AS2) or x and y properties (in AS3)

```
_visible/visible      false
```

-

0

0, ,

0

Modify -> Shape

Smooth, Straighten

Optimize

(

)

0

0

(gradient fills)

(gradient strokes)

(strokes)

0

0

0

API

API

Scaleform

EdgeAA

- , , .

- EdgeAA , /
/

DP .

2.3.3 비트맵

DP

(fill rate)

SWF [_alpha](#)(in AS2) or alpha

(in AS3) 0 [_visible](#) (in AS2) or [visible](#) (in AS3) [false](#) .

Scaleform [_alpha](#) 0

CPU .

[false](#) , CPU . SWF

(unloading) (reloading) [_visible](#) [false](#)

- / 2 16x32, 256x128, 1024x1024, 512x32

- JPEG

JPEG

gfxexport

DXT

SWF

GFX

(DXT)

gfxexport

-qp

-qh

DDS

(

.)

-

-

Shape

EdgeAA

-

loading

unloading

- SWF/GFX

SWF

JPG

SWF

1024 x

1024

JPEG

SWF

4MB

- UI

, 4

.(

4

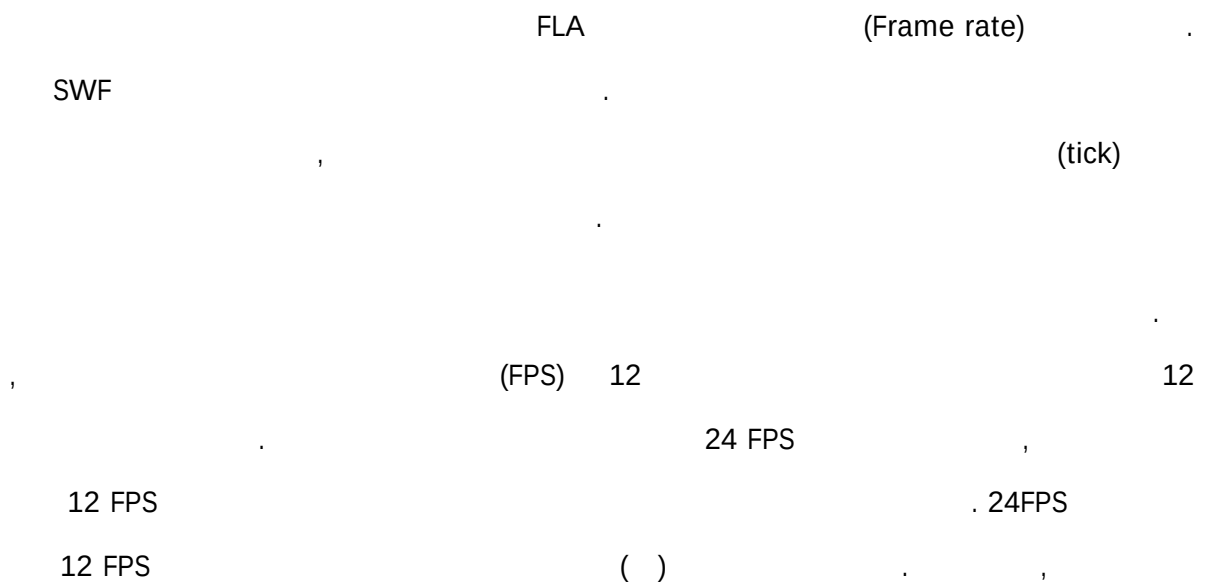
) gfxexport

-i DDS

. AMP

- , (fill rate), CPU
- gradient.swf
SWF import . gfxexport gradients.swf -d0
SWF
-
-
- PNG
-
-

2.3.4 애니메이션



5

5

주의: [onEnterFrame](#) (in AS2) or [Event.ENTER_FRAME](#) (in AS3)

[. onEnterFrame/EVENT.ENTER_FRAME](#)

[setInterval](#)

[. onEnterFrame/EVENT.ENTER_FRAME](#)

[setInterval](#)

[. 30~40FPS](#)

[, CPU](#)

[UI](#)

-

-

[on/off](#)

[\(attaching\)](#) [\(removing\)](#)

[_visible/visible = true/false](#)

-

- o

- o

[Tween](#)

- o [Scaleform](#)

[Tween](#)

[CLIK](#)

[Tween\(gfx.motion.Tween in AS2 or scaleform.clk.motion.Tween in AS3\)](#)

-

60

60 FPS

-

-

-

-

-

-

CPU

()

CPU

(, ,)

. 4

-

(translation)

(rotation)

2.3.5 텍스트와 폰트

-

SlotHeight

gfxexport

(48).

DP

. (

DP

.)

- `DP`
- `,`
- `, 30 FPS`
- `(TextField.variable )`.
- `“htmlText”` `. HTML`
- `( , )`
`gfxexport` `-fc, -fcl, -fcm` `“Font Overview”`
- `fontlib`
`. (“Font Overview” )`
- `,`
- `DP`
- `( )`
`DP`
- `,`

- (blur) (drop shadow) (knockOut)
- Scaleform UI DrawText API C++
- "DrawText API Reference"

3 액션 스크립트 최적화

3.1

- AS () ,
- AS
- AS
- attachMovie FLA
- AS
- “if”
- on() onClipEvent() onEnterFrame, onPress
- AS
- AS (onClipEvent, on)

(gotoAndPlay, play, stop)

-

gotoAndPlay/gotoAndStop

- o gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

- o gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

-

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

-

. C++

AS

[Gfx::Movie::SetVariableArray](#)

-

Scaleform

-

(Object)

```
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
    trace(fontArr[i]);
}
```

- ```
removeListener()
```

AS "if"

111

```

/ /
("com" "xxx",

```

yyy" ). Flash

c58923409876.functionName()

- AS SWF  
SWF
- AS
- SWF `trace()` , Publish Settings  
Omit Trace Actions  
trace
- 
- SWF AS ( : foo.bar.CustomClass)  
SWF  
SWF `delete`  
`delete foo.bar.CustomClass`
- 
- `onEnterFrames`
- (math)  
0  
Scaleform

### 3.1.1 루프

- 
- 
- 
- 

0

### 3.1.2 함수

- 
- with

### 3.1.3 변수/속성

- 
- “var” “var”

- 
- 

- null

SWF

null

- getter setter

## 3.2 *ActionScript 3*

AS3

### 3.2.1 엄격한 데이터 유형 설정

- 
- Object . Object AS3  
Object
- Array . Vector  
Array 4294967295( )  
Array
- Vector . Vector<\*>  
Vector<\*>  
Vector  
Vector<int>, Vector<String> Vector<YourFavoriteClass> Vector<\*>
- Object . flash.utils.Dictionary
- Gfx Flash
- (prototype) AS2  
AS3 . AS3

### 3.3 Advance

```

advance :

1. AS . onEnterFrame
 /Event.ENTER_FRAME .

2. . UI

3. . (_visible (AS2) or visible (AS3) true
 stop() . Advance

4. _global.noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3)
 . Advance

 "true" Advance (
)

 . _global.gfxExtensions (AS2) or scaleform.gfx.Extensions.enabled (AS3)
 true ()

5. . advance

6. .

```

### 3.4 *onEnterFrame* and *Event.ENTER\_FRAME*

*onEnterFrame*(AS2) or *Event.ENTER\_FRAME* (AS3)

*onEnterFrame*/*Event.ENTER\_FRAME*

- *setInterval*    *setTimeout*    *.setInterval*    :
- *clearInterval*
- *setInterval*    *setTimeout*    *onEnterFrame*/*Event.ENTER\_FRAME*  
    *onEnterFrame*/*Event.ENTER\_FRAME*

#### 3.4.1 *onEnterFrame* 정리

*onEnterFrame*    *delete*    :

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

*onEnterFrame*    *null*    *undefined*    . ( : *this.onEnterFrame* = *null*);

*onEnterFrame*    . *Scaleform*    *onEnterFrame*

### 3.5 *ActionScript 2*

#### 3.5.1 이벤트 상의 *onClipEvent*

*onClipEvent*()    *on*()    .    *onEnterFrame*, *onPress*    .

-

- onClipEvent on
  - this, \_global, arguments, super , 256

onLoad

onClipEvent(construct)

onEnterFrame

```
onClipEvent(construct)
{
 this.onLoad = function()
 {
 //function body
 }
}
```

onClipEvent(load)

### 3.5.2 Var 키워드

“var”

. “var”

. var

AS

```
var i = 1000;
countIt = function()
{
 num = 0;
 for(j=0; j<i; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}
```

```

:
var i = 1000;
countIt = function()
{
 var num = 0;
 var ii = i;
 for(var j=0; j<ii; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}

```

### 3.5.3 전캐싱(Precaching)

```

(var
)

```

```

function foo(var obj:Object)
{
 for (var i = 0; i < obj.size; i++)
 {
 obj.value[i] = obj.num1 * obj.num2;
 }
}

```

```

:
function foo(var obj:Object)
{
 var sz = obj.size;
 var n1 = obj.num1;
 var n2 = obj.num1;
 for (var i = 0; i < sz; i++)
 {
 obj.value[i] = n1*n2;
 }
}

```

```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
 // do something;
}
```

### 3.5.4 긴 경로의 전캐싱

```
mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;
```

```
var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;
```

### 3.5.5 복잡한 표현

C

```
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];
```

,

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;
```

```

for(i=0; i<3; i++)
{
 this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
 0x000000;
}

```

```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 _splqueue[slot_i].gosplash.text = _splstring;
 _splqueue[slot_i].gosplash2.text = splstring;
 _splqueue[slot_i].gosplash2.textColor = 0x000000;
}

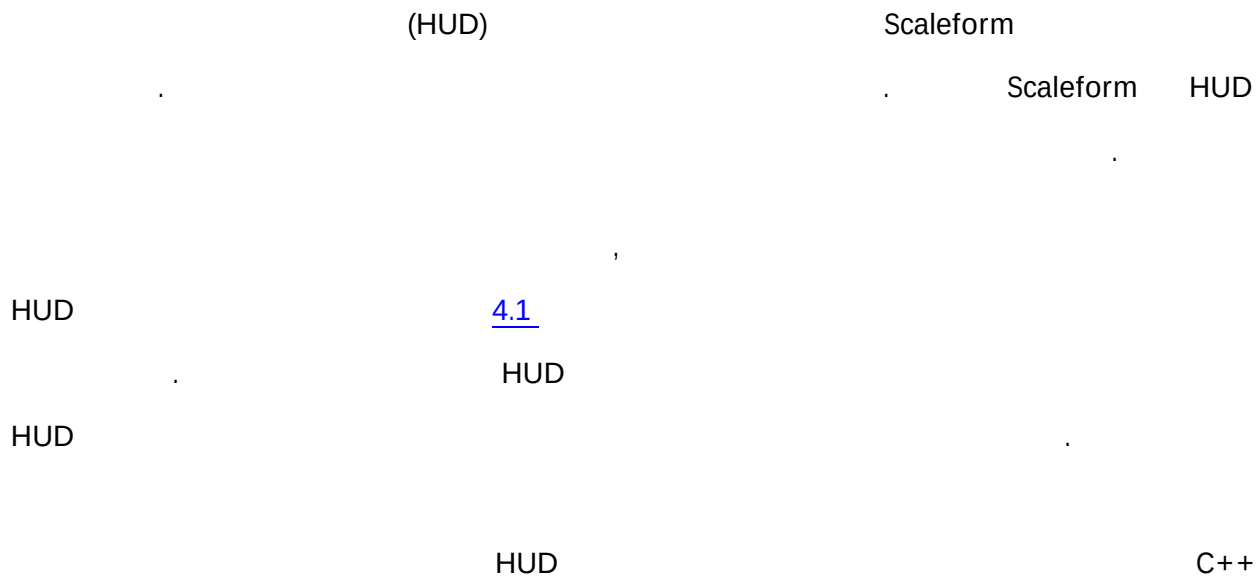
```

```

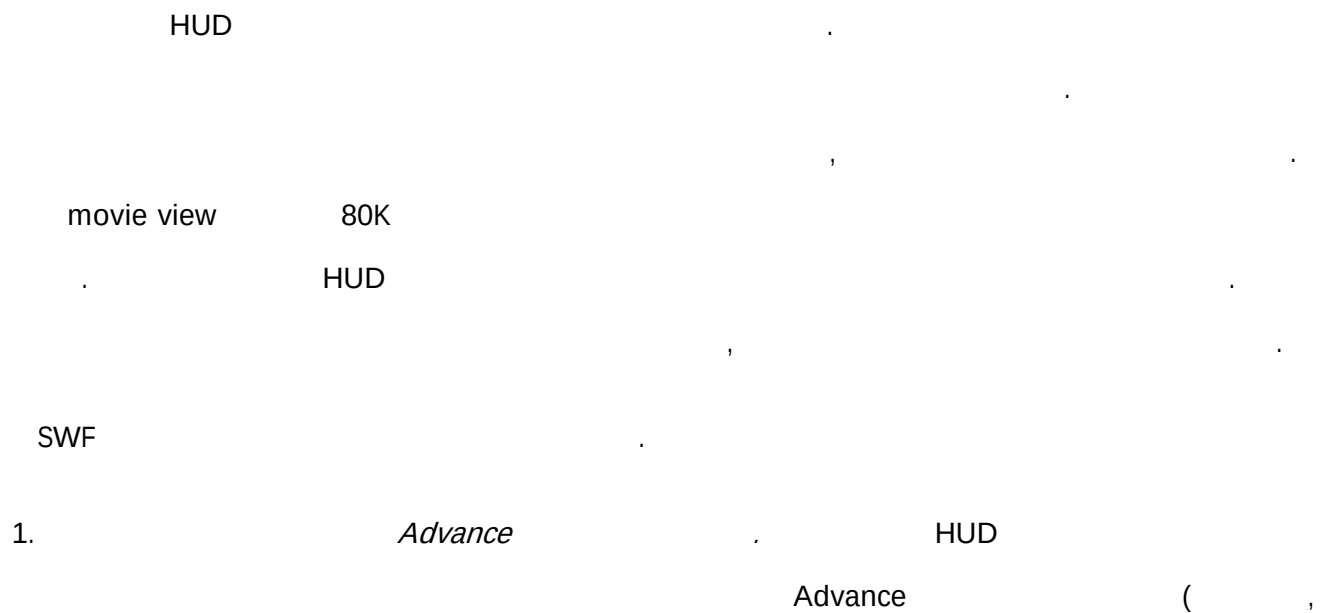
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 var elem = _splqueue[slot_i];
 elem.gosplash.text = _splstring;
 var gspl2 = elem.gosplash2;
 gspl2.text = splstring;
 gspl2.textColor = 0x000000;
}

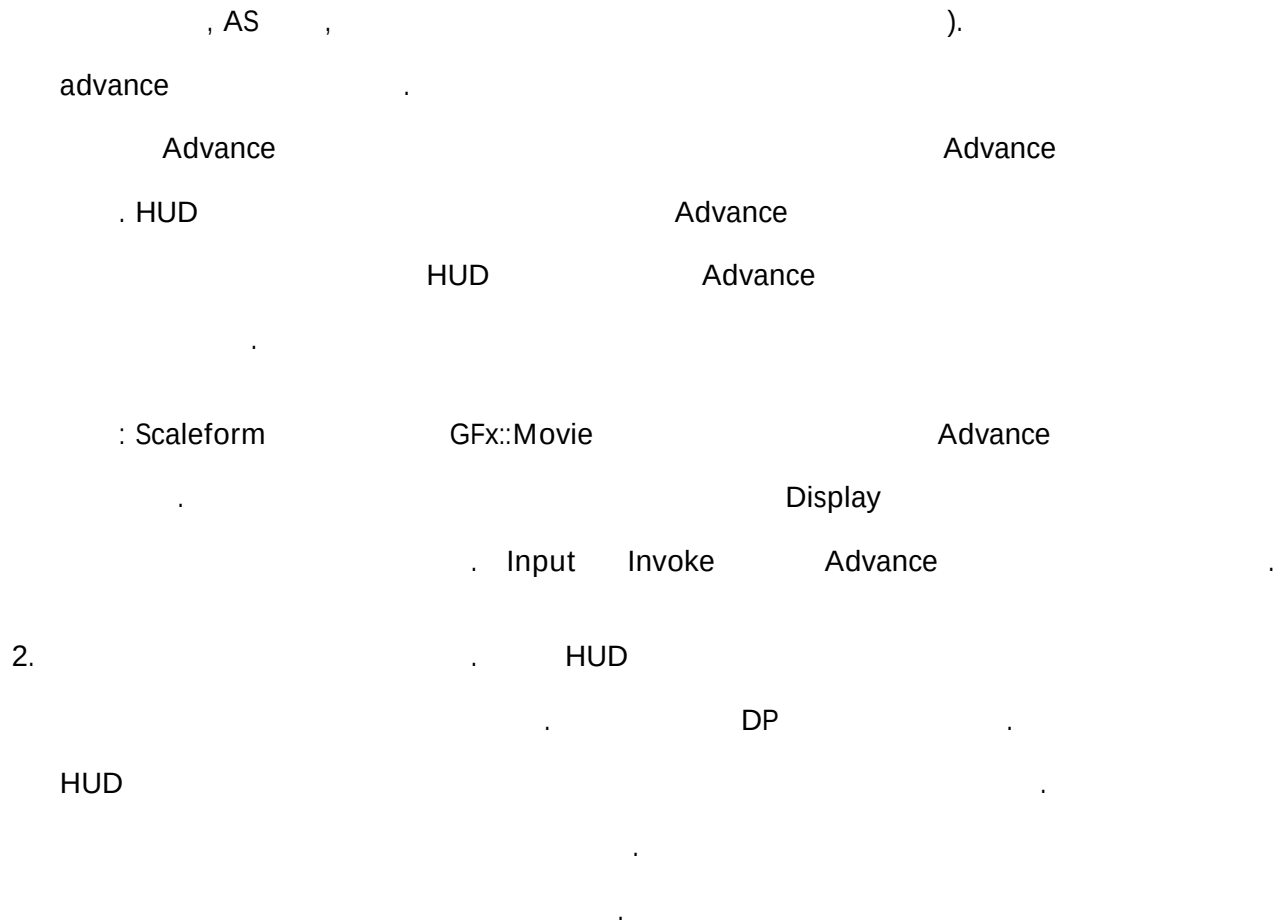
```

## 4 HUD 개발



### 4.1 SWF





## 4.2 SWF

loadMovie

1.

`_global.noInvisibleAdvance(AS2)` or `scaleform.gfx.Extensions.noInvisibleAdvance (AS3)`

`true` `_visible = false` (AS2) or `visible = false` (AS3) = `false` advance

`. HUD`

`. _visible = false`

```

 (
 Advance
).
 HUD
 (,
 ,
)
 HUD
 Advance
 . _global.gfxExtensions(AS2) or scaleform.gfx.Extensions.enabled (AS3) true

```

Scaleform

2. [SetVariableArray](#) Invoke

```

 (
)
 .
 SetVariableArray
 (
)
 Invoke

```

3. HUD [onEnterFrame\(AS2\)](#) or [Event.ENTER\\_FRAME \(AS3\)](#) . HUD  
[onEnterFrame/Event.ENTER\\_FRAME](#) Advance

4. AS  
 tick  
 AS Invoke  
 30-  
 40FPS 15-20FPS HUD

5. ,

## 4.3

```

(
)

```

. Flash vs C++

DP

. 4.2

1.

(Border)

(Frame)

Scaleform

. HUD

C++

2.

C++

Render::Renderer

RenderString ID

. C++

AS

## 4.4 ( )

HUD

. 4.3

1. HUD

Advance

, Invoke

( ) HUD

Advance

.(

Advance/invoke )

CPU

2.

(custom)

, Render::Renderer

(override)

C++

C++

. Render::Renderer

(custom)

, HUD  
 GfX  
 3. Render::Renderer  
 HUD Advance  
 : Scaleform- 3

## 4.5 HUD

HUD C++  
 . Scaleform HUD  
 Scaleform  
 advance HUD  
 Scaleform Player  
 C++ Scaleform  
 DP  
 Scaleform  
 Scaleform /  
 Scaleform DrawText API  
 C++  
 Scaleform UI  
 FAQ , [Font](#)  
[Overview](#) , [DrawText API Reference](#)

## HUD

- (nest)
- 
- FAQ ["Graphics Rendering & Special Effects"](#)
- PC Wii™ (
- .) :
- o Gfx::Movie::SetMouseCursorCount(0);
- o
- HUD
- noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3) [\\_visible](#)  
(AS2) or visible (AS3)= false
- Invoke . 2 (
- ) Invoke . ( )
- FAQ [section](#)
- [2.1](#) ["Art & Assets"](#)

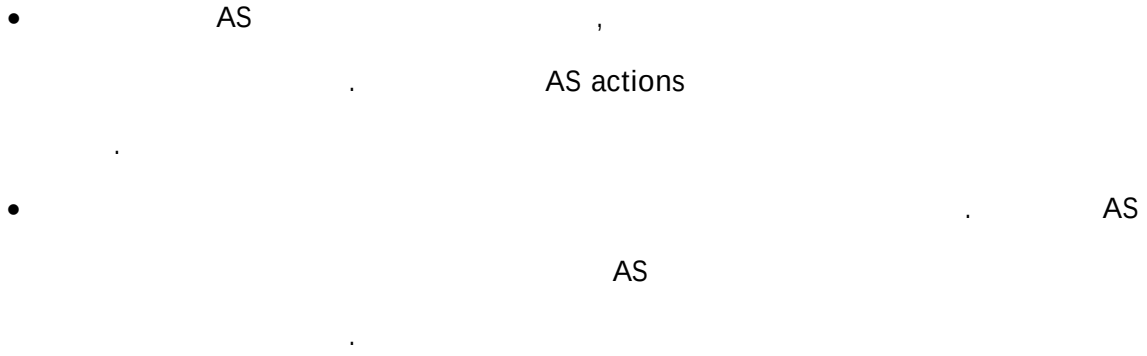
## 5 일반적인 최적화 팁

### 5.1

- FLA
- (loop) FPS
- 
- 
- 
- / AS
- (Layer 1, Layer 2 )

### 5.2

- ,
- 
- `trace()` . ( )



## 6 추가문서

ActionScript 2.0 Best Practices

[http://www.adobe.com/devnet/flash/articles/as\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/as_bestpractices.html)

Flash 8 Best Practices

[http://www.adobe.com/devnet/flash/articles/flash8\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html)

Flash ActionScript 2.0

[http://www.adobe.com/devnet/flash/articles/actionscript\\_guide.html](http://www.adobe.com/devnet/flash/articles/actionscript_guide.html)